Therefore, $\dfrac{1+\cos 2\theta + i\sin 2\theta}{1+\cos 2\theta - i\sin 2\theta} = \dfrac{1+z}{1+\dfrac{1}{z}} = \dfrac{(1+z)z}{z+1} = z$.

Therefore, $\left(\dfrac{1+\cos 2\theta + i\sin 2\theta}{1+\cos 2\theta - i\sin 2\theta}\right)^{30} = z^{30} = (\cos 2\theta + i\sin 2\theta)^{30}$

$= \cos 60\theta + i\sin 60\theta$.

Example 2.31

Simplify (i) $(1+i)^{18}$ (ii) $(-\sqrt{3}+3i)^{31}$.

Solution

(i) $(1+i)^{18}$

Let $1+i = r(\cos\theta + i\sin\theta)$. Then, we get

$r = \sqrt{1^2+1^2} = \sqrt{2}$; $\alpha = \tan^{-1}\left(\dfrac{1}{1}\right) = \dfrac{\pi}{4}$,

$\theta = \alpha = \dfrac{\pi}{4}$ ($\because 1+i$ lies in the first Quadrant)

Therefore $1+i = \sqrt{2}\left(\cos\dfrac{\pi}{4} + i\sin\dfrac{\pi}{4}\right)$

Raising to power 18 on both sides,

$$(1+i)^{18} = \left[\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)\right]^{18} = \sqrt{2}^{18}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)^{18}.$$

By de Moivre's theorem,

$$(1+i)^{18} = 2^9\left(\cos\frac{18\pi}{4} + i\sin\frac{18\pi}{4}\right)$$

$$= 2^9\left(\cos\left(4\pi+\frac{\pi}{2}\right) + i\sin\left(4\pi+\frac{\pi}{2}\right)\right) = 2^9\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$$

$(1+i)^{18} = 2^9(i) = 512i$.

(ii) $(-\sqrt{3}+3i)^{31}$

Let $-\sqrt{3}+3i = r(\cos\theta + i\sin\theta)$. Then, we get

$r = \sqrt{\left(-\sqrt{3}\right)^2 + 3^2} = \sqrt{12} = 2\sqrt{3}$,

$\alpha = \tan^{-1}\left|\dfrac{3}{-\sqrt{3}}\right| = \tan^{-1}\sqrt{3} = \dfrac{\pi}{3}$,

$\theta = \pi - \alpha = \pi - \dfrac{\pi}{3} = \dfrac{2\pi}{3}$ ($\because -\sqrt{3}+3i$ lies in II Quadrant)

Therefore, $-\sqrt{3}+3i = 2\sqrt{3}\left(\cos\dfrac{2\pi}{3} + i\sin\dfrac{2\pi}{3}\right)$.

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